Weiqing Chen

Machine Learning Researcher | Multimodal AI & Computational Biology

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PROFILE

Machine-learning researcher developing multimodal foundation models and research tools for computational pathology, single-cell biology, spatial biology, dynamical modeling of biological systems, and multi-omics analysis, including transcriptomics, proteomics, and metabolomics. Led end-to-end development of **OmiCLIP, a visual-omics foundation model, and its downstream platform, Loki** (**Nature Methods**), from model design and data curation through training, evaluation, and software release, including downstream task development and applications to biological research. International and interdisciplinary background: trained in computer science in China, theoretical biology in Japan, and AI for biomedicine in the United States. First and co-first author of work published in **Nature Methods**, **Nature Communications**, and **Circulation Research**, with a first-author manuscript in third-round revision at **Nature**.

TECHNICAL EXPERTISE

Programming & Frameworks: Python (PyTorch, Hugging Face, scikit-learn, NumPy/pandas, AnnData/Scanpy, Squidpy), R (Seurat, DESeq2), MATLAB; Linux, HPC (Slurm), Git, Docker, AWS; cloud workflow deployment.
Machine Learning: Foundation models, multimodal representation learning, contrastive (CLIP-style) vision–language pretraining, vision transformers, transfer learning and fine-tuning, zero-shot retrieval and linear probing, Bayesian modeling, classical machine learning (logistic regression, random forests, SVMs), multi-GPU training on HPC, production deployment of foundation models.
Computational Biology: Computational pathology, transcriptomics, proteomics, metabolomics, spatial omics (10x Visium / Visium HD / Xenium / MERFISH / Stereo-seq / CODEX), single-cell omics, multi-omics integration, dynamical modeling of biological systems.
Imaging & Data: OpenSlide, OpenCV, QuPath, Cellpose, OME-Zarr; H&E, IHC, mIF, and CODEX image analysis; whole-slide images and TB-scale pathology datasets.
Languages: Chinese (native), English, Japanese (JLPT N1).

EDUCATION

Cornell University, Weill Cornell Graduate School of Medical Sciences Aug. 2022 – 2027 (Expected)
Ph.D. Candidate, Physiology, Biophysics and Systems Biology | Advisor: Guangyu Wang, Ph.D.

Kyoto University Oct. 2019 – Sep. 2021
Master of Science, Systemic Life Science | Advisor: Naoki Honda, Ph.D.

Nanjing University of Posts and Telecommunications Sep. 2015 – Jun. 2019
Bachelor of Computer Science and Technology

RESEARCH & ENGINEERING EXPERIENCE

Houston Methodist Research Institute / Weill Cornell Medicine Aug. 2022 – Present
Visiting Graduate Research Fellow | Houston, TX

- Led **OmiCLIP**, a visual-omics foundation model, from model design and data curation through training and evaluation, aligning histopathology with transcriptomics via contrastive learning on **2.2 million paired tissue patches** from **1,007 samples across 32 organs**, curated from **113 studies**. Benchmarked the model and its downstream platform, Loki, against **22 state-of-the-art methods across five core downstream task categories**; co-first author, **Nature Methods** (2025).
- Built and released **Loki**, the open-source downstream platform that translates OmiCLIP into five capabilities for biological research: tissue alignment, molecular annotation, cell-type decomposition, image-transcriptomics retrieval, and spatial gene-expression prediction from H&E images. Across visual-omics projects and Loki2, worked with **TB-scale pathology datasets spanning over 100,000 images**.
- Led end-to-end development of a **single-cell pathology foundation model** for molecular decoding of H&E images, spanning model design, data curation, training, evaluation, software development, downstream task development, and applications to biological research; first-author manuscript in **third-round revision at Nature**.
- Led downstream application design and analysis for **Thor**, integrating histology and spatial transcriptomics at cellular resolution to enable CNV-based tumor mapping in ductal carcinoma in situ and identify vascular PLA2G2A enrichment in post-LVAD heart tissues; co-first author, **Nature Communications** (2025).

RESEARCH & ENGINEERING EXPERIENCE (CONTINUED)

Houston Methodist Research Institute / Weill Cornell Medicine *(continued)*

- **Collaborative biological discovery:** Computational analyses with experimental teams linked LATS1/2-CD38 signaling to endothelial senescence and thrombosis, mTORC1 and AIBP-VEGFR3 to lymphatic development, and metastatic precursor cells to candidate HSP90AA1/HSP90AB1 biomarkers; co-first author, **Circulation Research**; contributing author, **Developmental Cell** and **Nature Communications**.
- **Technical mentoring and training:** Mentored **five** junior Ph.D. students, rotation students, and research assistants through shared pipelines, code review, and result reproduction; trainee analyses supported downstream applications and one trainee built an independent pipeline. Contributed materials and training for the **ISMB 2026 tutorial** on visual-omics foundation models; coordinated compute needs with HPC platform providers.

Houston Methodist Research Institute Sep. 2021 – Jun. 2022
Research Assistant Intern | Remote

- Analyzed single-cell data and evaluated model robustness for **cellDancer**, a deep-learning method for cell-dependent RNA velocity and cell-state dynamics; contributing author, **Nature Biotechnology**.

Kyoto University Apr. 2020 – Mar. 2021
Research Assistant | Kyoto, Japan

- Developed a Bayesian active-inference framework for gradient sensing in goal-directed migration, connecting theoretical biology with computational modeling; presented the work at national biophysics and mathematical biology meetings in Japan.

Gausscode AI Automotive Solution Sep. 2018 – Sep. 2019
Software Engineer and Data Analyst Intern | Nanjing, China

- Built data-collection pipelines and analysis workflows for an AI and big-data company serving the automotive industry, and contributed to the development of its internal data platform.

SELECTED PUBLICATIONS

[*]Co-first authors

1. **W. Chen**[*], Q. Li[*], H. Cheng[*], et al. A single-cell pathology foundation model for molecular decoding of H&E images. Under revision in **Nature**, third round.
2. **W. Chen**[*], P. Zhang[*], et al. A visual-omics foundation model to bridge histopathology with spatial transcriptomics. **Nature Methods**, 2025.
3. P. Zhang[*], **W. Chen**[*], et al. Thor: a platform for cell-level investigation of spatial transcriptomics and histology. **Nature Communications**, 2025.
4. S. Kotla[*], J. Lee[*], K. A. Ko[*], **W. Chen**[*], et al. LATS1/2-CD38 metabolic rewiring links senescence to intraplaque thrombosis. **Circulation Research**, 2026.
5. S. Li[*], P. Zhang[*], **W. Chen**, et al. A relay velocity model infers cell-dependent RNA velocity. **Nature Biotechnology**, 2024.
6. X. Wang, M. Duan, A. J. Snyder, …, **W. Chen**, et al. Deep-learning-enabled multi-omics analyses for prediction of future metastasis in cancer. **Nature Communications**, 2026.
7. F. Han, S. Simeroth, J. Zhu, I. Gryniuk, A. Pranay, **W. Chen**, et al. Lymphatic endothelial mTORC1 instructs metabolic and developmental signaling during lymphangiogenesis. **Developmental Cell**, 2025.
8. J. Kim, S. Chaudhary, **W. Chen**, et al. APOA1 binding protein promotes lymphatic cell fate and lymphangiogenesis by relieving caveolae-mediated inhibition of VEGFR3 signaling. **Nature Communications**, 2025.

SELECTED PRESENTATIONS & AWARDS

ISMB 2026 | Tutorial on visual-omics foundation models; contributed materials and training. Jul. 2026
ICIBM 2025 | Presentation on OmiCLIP, a visual-omics foundation model. Aug. 2025
Texas Single Cell & Spatial Genomics Seminar Series | Presentation. Nov. 2024
DeBakey Heart and Vascular Center Research Day | Poster; Best Poster Award. May 2024
8th Annual Cardiovascular Bioengineering Symposium | Poster. May 2024

Awards & honors: Best Poster Award, PBSB Program Annual Retreat, Weill Cornell Medicine (Apr. 2026); JASSO Honors Scholarship, Japan Student Services Organization (2019, 2020); NJUPT: Academic Excellence Award (2018), Excellent Student Cadre and Social Work Excellence Awards (2017–2019), Vice Chairman of the Student Association for Science and Technology (2017–2018).